

Differentiate $x^{(\ln x)^{(\ln x)}}$

Complete the following.

$$\frac{d}{dx} \left[x^{(\ln x)^{\ln x}} \right] \Big|_{x=e}$$

Solution:

Let $u = x^{(\ln x)^{\ln x}}$. Then, we have

$$\begin{aligned} \ln u &= \ln \left[x^{(\ln x)^{\ln x}} \right] \\ &= (\ln x)^{\ln x} (\ln x) \end{aligned}$$

Let $v = (\ln x)^{\ln x}$. Then, we have

$$\begin{aligned} \ln v &= \ln [(\ln x)^{\ln x}] \\ &= (\ln x) \ln(\ln x) \end{aligned}$$

Differentiate both sides:

$$\begin{aligned} \frac{1}{v} v' &= \frac{d}{dx} [(\ln x) \ln(\ln x)] \\ &= \frac{1}{x} \ln(\ln x) + (\ln x) \left(\frac{1}{\ln x} \cdot \frac{1}{x} \right) \\ &= \frac{\ln(\ln x)}{x} + \frac{1}{x} \end{aligned}$$

Thus, we have

$$\begin{aligned} v' &= v \frac{\ln(\ln x) + 1}{x} \\ &= (\ln x)^{\ln x} \frac{\ln(\ln x) + 1}{x} \end{aligned}$$

Now return to u :

$$\ln u = v \ln x.$$

Differentiate both sides:

$$\frac{1}{u} u' = v' \ln x + v \frac{1}{x}$$

So, we have

$$u' = u \left(v' \ln x + \frac{v}{x} \right).$$

Evaluate at $x = e$:

$$\ln e = 1, \quad v = (1)^1 = 1, \quad v' = 1 \cdot \frac{0+1}{e} = \frac{1}{e}, \quad u = e^1 = e.$$

Hence,

$$\begin{aligned} u' &= e \left(\frac{1}{e} \cdot 1 + \frac{1}{e} \right) \\ &= e \left(\frac{2}{e} \right) \\ &= 2 \end{aligned}$$